

4(a)	replacing cells that are, damaged / destroyed / worn out / old ;	1
4(b)	any two from: result of mutation ; A example proto-oncogene to oncogene tumour suppressor gene inactive I mutation / mutated, <i>as a single term</i> short(er) / fast(er) / many / continuous, cell cycle(s) / cell divisions ; A shorter interphase A cells do not stop dividing I ref. to mitosis no contact inhibition / mitosis continues beyond space available / cell formation may spread to other nearby areas / AW ; ref. to normal (cell cycle) checkpoints not occurring ; A <i>idea that</i> substances needed for error checking not, available / functioning A errors not checked loss of (original / normal) function ; A example A non-functional (cell cycle continues because) fault in / errors in / no response to, cell signalling ; cells do not carry out apoptosis / no programmed cell death ; AVP ; e.g. different cell metabolism increase in telomerase (activity) telomeres do not shorten as quickly formation / presence, of blood vessels (feature of tumour) A angiogenesis lack of adhesion between cells ref. metastasis / described e.g. cells travel in, blood / lymph, to form tumours elsewhere in body I ref. to tumour spread in blood	2

Question	Answer	Marks
4(c)	any two from: <i>idea that</i> after each, cell cycle / cell division / DNA replication, telomeres shorten ; (long telomeres means that DNA) replication can take place more times ; A allows continuous (DNA) replication <i>context is DNA and not cell replication</i> <i>idea that</i> <u>ends</u> of chromosomes are protected ; telomeres do not contain, genes / genetic information ; A telomeres are non-coding detail ; e.g. <i>idea that</i> after replication nucleotides at ends of DNA are lost lagging strand replication leads to unpaired nucleotides DNA polymerase cannot continue to end on lagging strand (replication) (long telomeres) allow, <u>long</u> life span / many divisions to occur / many mitoses ; AVP ; e.g. prevents chromosome ends from joining otherwise, mitosis could not occur / may lead to cell death	2
4(d)	any three from: differentiation has already started / AW ; A they are not undifferentiated A GMP cells are, differentiated / specialised (compared to stem cells) no self-renewal / AW ; A cannot produce a stem cell when they divide cannot form, all blood cell types / correct named ; (only) forms (immature) neutrophils and monocytes ;	3
4(e)	macrophage/s ; I mature monocytes / granulocyte / phagocyte	1

Question	Answer	Marks
4(f)	antibody / immunoglobulin ;	1
4(g)	<i>mp2 and 3 need to see knowledge that self-antigens are on, self- / body / own, cells</i> any three from: 1 <i>idea that, next stage / after thymus, is (T-lymphocyte) release into general circulation / blood ;</i> 2 (if released) exposure to / activation by, (self-antigens on body) cells or (on exposure to self-antigens), immune response occurs against (body) cells ; ora only (T)lymphocytes that will respond only to, pathogens / foreign substances, are, left / released 3 (if released) harm to / destruction of / kills, (body) cells ; 4 need to prevent / otherwise, formation of memory T-cells ; 5 AVP ; e.g. need to prevent / (if released) may cause, autoimmune, disease / response (otherwise) T-helper will release cytokines to, enhance / AW, immune response against (body) cells (if cytokine released) may lead to, phagocytosis of / antibodies produced against, (body) cells (otherwise) T-killer cell will release substance to kill cells <i>also mp3</i> (faulty T-lymphocytes) cannot bind to, foreign / non-self, antigens	3